

Matgeo-2.6.30

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Question

If the area of a triangle is 35 square units with vertices $(2, 6)$, $(5, -4)$ and $(k, 4)$, then find k .

Solution

$$\vec{A} = \begin{pmatrix} 2 \\ 6 \end{pmatrix}, \quad \vec{B} = \begin{pmatrix} 5 \\ -4 \end{pmatrix}, \quad \vec{C} = \begin{pmatrix} k \\ 4 \end{pmatrix} \quad (1)$$

$$\vec{B} - \vec{A} = \begin{pmatrix} 3 \\ -10 \end{pmatrix}, \quad \vec{C} - \vec{A} = \begin{pmatrix} k-2 \\ -2 \end{pmatrix} \quad (2)$$

$$\text{Area} = \frac{1}{2} \left| (\vec{B} - \vec{A}) \times (\vec{C} - \vec{A}) \right|$$

$$35 = \frac{1}{2} \left| \begin{pmatrix} 3 \\ -10 \end{pmatrix} \times \begin{pmatrix} k-2 \\ -2 \end{pmatrix} \right|$$

Solution (Continuation)

$$35 = \frac{1}{2} |3(-2) - (-10)(k - 2)| \quad (3)$$

$$= \frac{1}{2} |-6 + 10k - 20| \quad (4)$$

$$= \frac{1}{2} |10k - 26| \quad (5)$$

Thus, $|10k - 26| = 70$

$$10k - 26 = 70 \quad \Rightarrow k = \frac{48}{5} \quad (6)$$

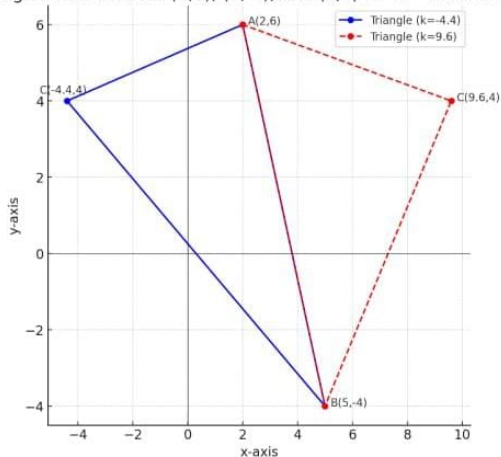
$$10k - 26 = -70 \quad \Rightarrow k = -\frac{22}{5} \quad (7)$$

Conclusion

Conclusion: The possible values of k are $k = \frac{48}{5}$ or $k = -\frac{22}{5}$.

Graphical Representation

Triangles with vertices $(2,6)$, $(5,-4)$, and $(k,4)$ for $k=-4.4$ and $k=9.6$



Figure